1) CO1 AT - 1.

a) MEET AT NOON.

Using Caesar cipher Each letter upon position

Original letter	Shift + 1	Encrypted letter
M	N → O → P → Q	Q
E	F → G → H → I	I
E	F → G → H → I	I
T	U → V → W → X	X
A	B → C → D → E	E
T	U → V → W → X	X
N	O → P → Q → R	R
O	P → Q → R → S	S
O	P → Q → R → S	S
N	O → P → Q → R	R

Encrypted : ~~Q I I X~~

Encrypted message : Q I I X E X R S S R.

b) Q → M

I → E

X → T

Authorized member. Shifts each encrypted. Letter four positions backward in the Alphabet.

MEET AT NOON ⟶ QIIX EX RSSR.

2).

a). The most Likely original message is

THIS IS. A SECRET.

message Appears to use the Atbash cipher which Replaces each Letter with its opposite Letter in the Alphabet.

b). A is replaced By Z, B by Y & C by X.

Encrypted Letter	Original Letter.
G	T
~~B~~ S	H
R	I
H	S
Z	A.

HXVIG SECRET.

SECRET ⟶ HXVIG.

a) Vigenere cipher. uses a repeating keyword. to Enaypt each letter.

Plaintext: HELLO.

Keyword: KEYKE.

$$C = (P + K) \bmod 26$$

H	K	(7+10) mod 26 = 17	R.
E	E	(4+4) mod 26 = 8.	I
L	Y.	(11+24) mod 26 = 9	J.
L	K	(11+10) mod 26 = 21	V
O	E	(14+4) mod 26 = 18	S

b).

- Key is repeated to match. the message length.
- Plaintext letter is combined with its corresponding.
- Keyword letter
- Different Keyword Letters produce different shifts in Alphabet..
- Plaintext Letter when paired with different Key letter
- recievers uses the same keyword to decrypt the ciphertext

4).

b)

Encrypt the Secret message Before embedding it. AES use to remain the unreadable without the Decryption Key.

use keyed Steganograph: data using Secret key and embedding pattern to make difficult for unauthorized users.

4).

A). Metadata Analysis: image property file detail & metadata for sapicious information.

B) File Size Analysis: image File size is usually. Large for its Dimension and Quality.

C). Pixel Analysis: Pixel Value and Colon channels

D) LSB Analysis: least significants bits of pixel.

E) statistical Analysis: compare pixel value distribution to identify possible

F). Steganography: To identify and Extract Hidden data when the Embedding Method is known.

5)

A)

Ciphertext :

WERCL TEERD SOEE FE AO CAI VDEN.

WE ARE DISCOVERED FLEE AT ONCE

This phrase is a Well Known Ex of Transposition.

b)

1. Preserve letter frequency

2. Limited Security.

3. Susceptible to attacks.

4. No modern Security Guarantees.

5. Key recovery is Easier.

c).

Replace AES 256 Authorized. Authentication Encryption Such as AES-GCM.

Benefits:

- provide Strong confidentiality.
- Detects unauthorized modification of Encrypted message.
- Cryptographic Keys rather than Relying on Simple Letter rearrangement.
- Suitable for protecting Sensitive corporate data. When implemented correctly with Secures key management & unique nonces.